

TEXAS A&M UNIVERSITY
ROBOTICS TEAM & LEADERSHIP EXPERIENCE

DRON Design Review

Project Lead: Ian [2] Wilhite

Members: Aidan Briggs, Asbah Talal, Charlie Weatherspoon, Drew Wheaton, Chris Ambroziak, Jackson Shouba, Luke Breedlove, Matthew Shi, Om Bhatt, Owen Thomas, Riya Shah, Ryo Kato, Visith So

Project Objectives



ID	Description
PO-1	DRON shall collect useful information concerning structural fires.
PO-2	DRON shall relay gathered information to first responders.
PO-3	DRON shall be controlled remotely or autonomously.
PO-4	DRON shall fail to a safe state.
PO-5	DRON shall remain within allowable safety standards for all phases of flight.
PO-6	DRON shall sustain flight for 10 minutes.

Mechanical



What is included in the mechanical section depends on what phase you are in. Include some screenshots of crucial CAD or drawings that needs to be discussed, but make sure to upload everything you have into the drive. Some examples of what to include are:

- Sketches (whiteboard, paper, etc.)
 - Mainly for Research Phase projects to include this before CAD is done
- CAD
 - More geared towards Prototyping Phase projects, include as much as is done
- Simulations (FEA, fluid, etc.)
- Any calculations that support design choices
 - Solid mechanics calculations for example

Preliminary Mechanical System Requirements



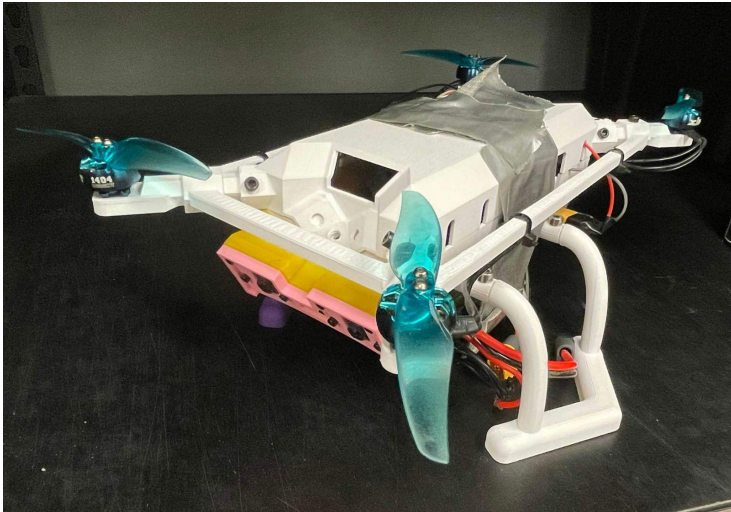
ID	Description	Source	Derivation
M-R-1	The chassis shall withstand a free fall of 10 meters.	PO-4	The drone may fly as high as a 3-story building. In event of power failure, the drone may experience free fall.
M-R-2	The chassis shall maintain structural integrity during all phases of operation.	PO-5	In order to ensure safe operations, no component may fail from loads.
M-R-3	The chassis shall have a factor of safety of 1.5 for all expected forces and moments	PO-5	The chassis is expected to experience loads in many directions during operation. A factor of safety is necessary to insure structural stability.
M-R-3.1	The arm shall be capable of withstanding a moment of TBD Nm in the vertical direction	PO-5	Expected moment produced by downwards force from motors.
M-R-3.2	The arm shall be capable of withstanding a moment of TBD Nm in the horizontal direction	PO-5	Expected moment produced by rotational force from motors.
M-R-4	The chassis shall maintain a factor of safety of 1.5 for exposure to extreme temperatures experienced near a fire.	PO-5	In order to ensure safe operations, no component may fail in proximity from a fire.
M-R-5	The chassis shall comfortably and securely support all critical components.	PO-5	Unsecured components may cause failure during flight operation.
M-R-6	All electrical components must be easily accessible.	PO-5	Ease of repair is critical to maintaining safety standards.

Two Drones in Parallel



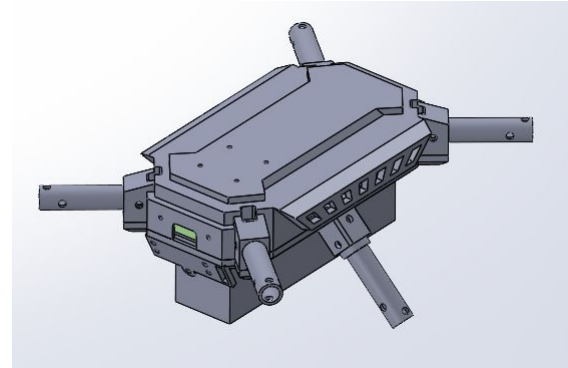
“Duck Tape” Test Drone

- Purely for verification testing by autonomy and visualization subteams
- Easily producible, repairable, modifiable
 - Full assembly within few days



V4 Production Drone

- Product-like final design
- Robust and reliable mechanical design that meets all system requirements
- Supported by thorough research and testing

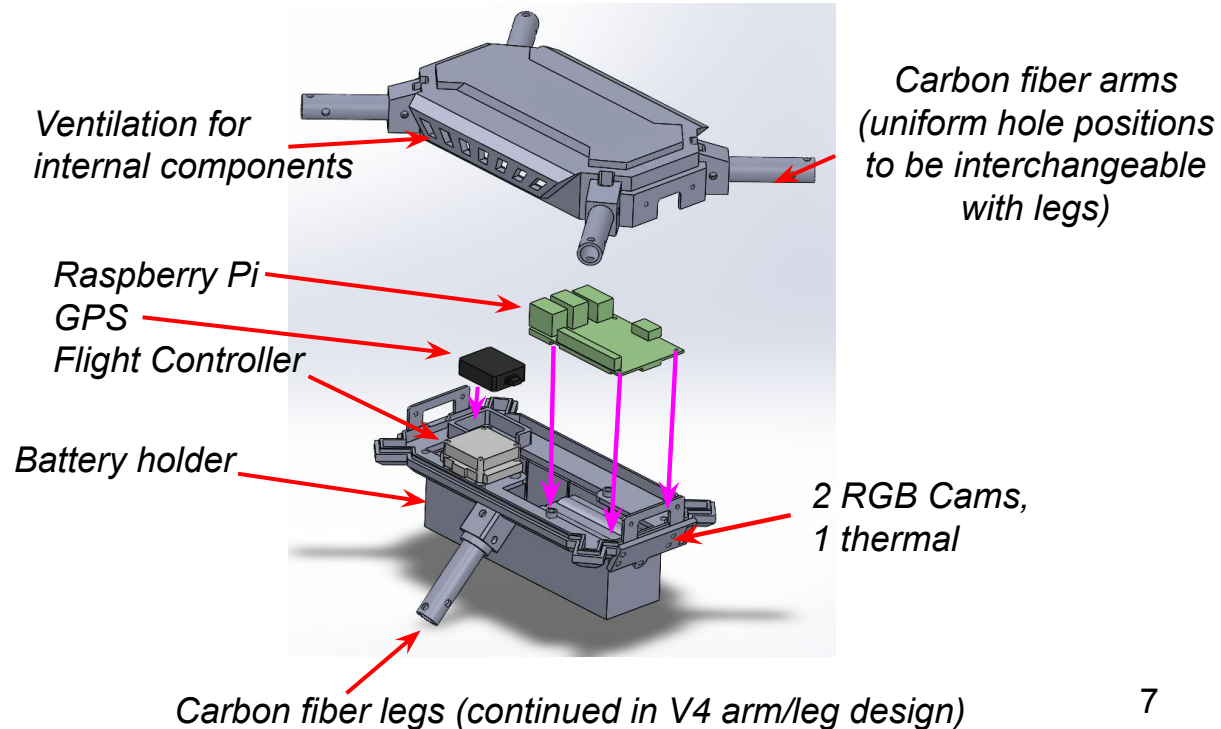


V4 Chassis Design



Improvements:

- More space for components
 - Less wire tangling
- Sturdier frame
- Easier access to batteries
- More presentable design



V4 Chassis Design

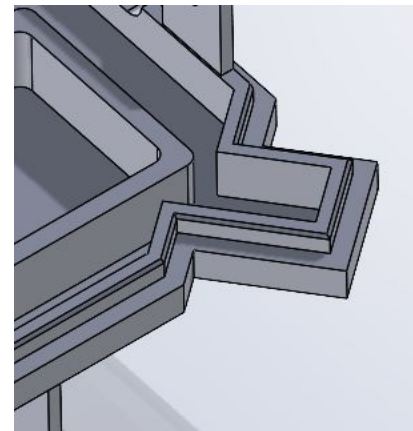
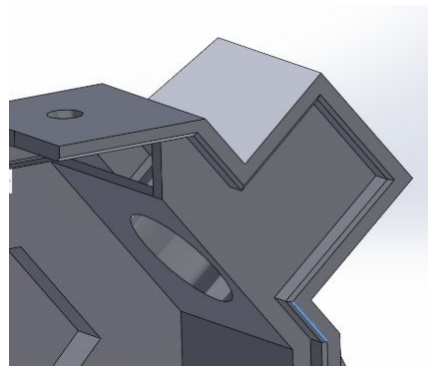


Other features (vs test drone)

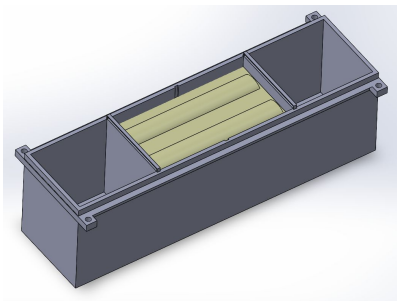
- Configurable, accessible battery slots
- Lip and Groove interlock for top and bottom halves of chassis

Remaining Features to be added:

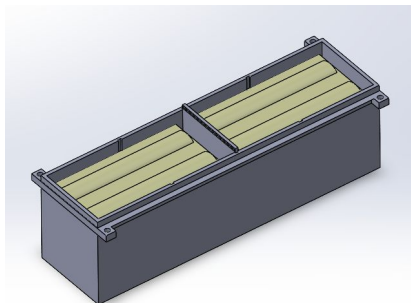
- Improved camera and thermal camera mount
- Clamps to attach arms if drilled holes do not work



Lip and Groove for easy assembly



1 battery



2 battery

V4 Arm/Leg Design



Beating Stick (**Carbon Fiber Tubes**)

- 16mm OD 2mm thickness
- 3k plain weave mattle
- For arms and legs
- ~~+10% speed buff~~



Manufacturing

- Dremel cutting
- Diamond cutting wheel
- PPE
 - masks
 - safety glasses
 - **Not** not in lab

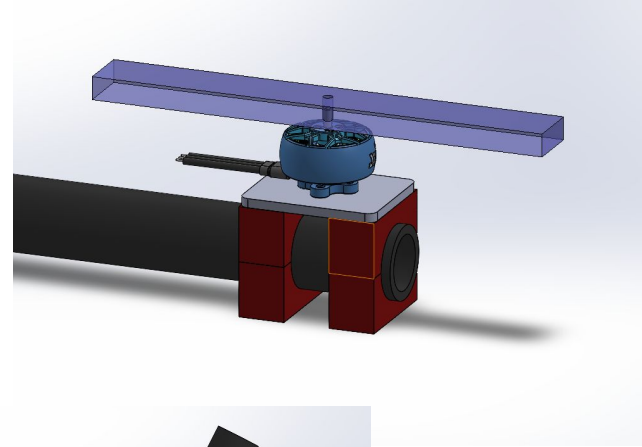


V4 Arms



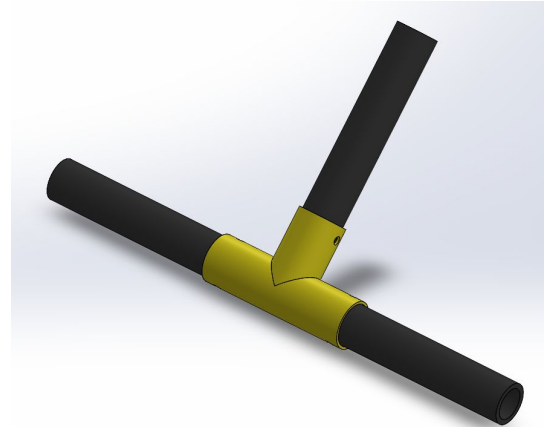
Arm

- Bolted arm into frame
- Clamped motor mount
- Bolted motor into mount
- 10 cm tube
 - (2 cm into frame, motor ~8cm from frame)
- All CF holes are M3

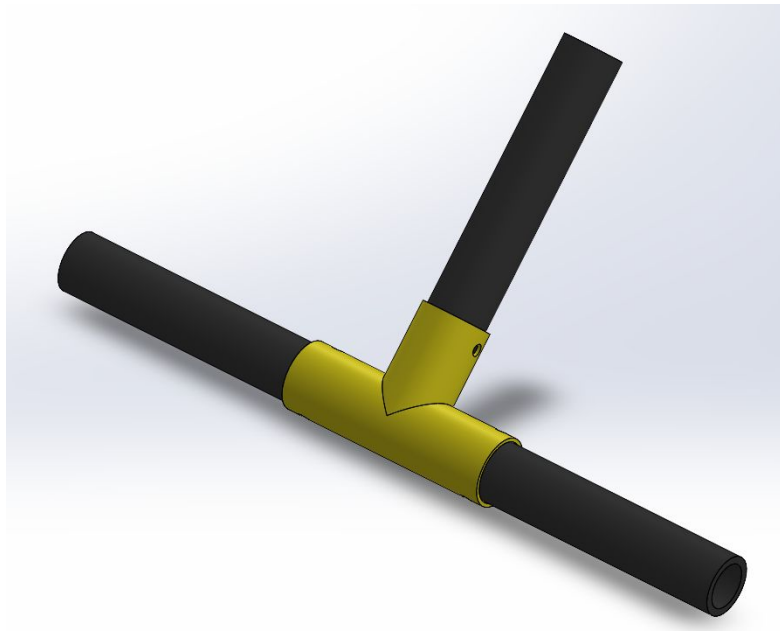


Legs

- 2 carbon fiber tubes
- 2 bolts in each tube
- 20 & 10 cm tubes



V4 Legs



- 2 carbon fiber tubes
- 2 bolts in each tube
- 20 & 10 cm tubes

V4 Finite Element Analysis



When drop testing one beam longitudinally from 30m, leg experiences distributed stress of around 47 MPa (**Figure 1**).

When drop testing one T-shaped foot of the drone from 30m, leg experiences small, concentrated high stress of around 119 MPa (**Figure 2**).

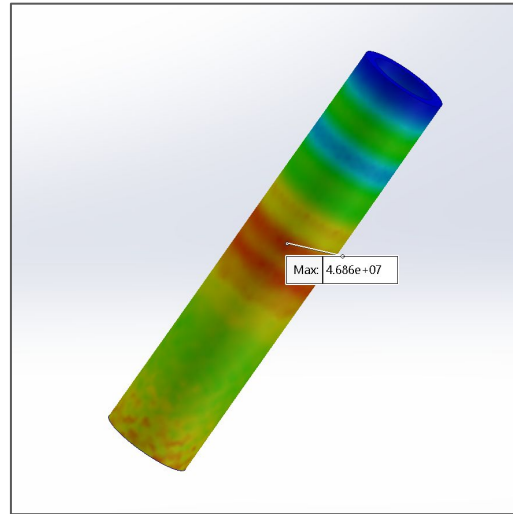


Figure 1. Drop test on a single pipe, longitudinally

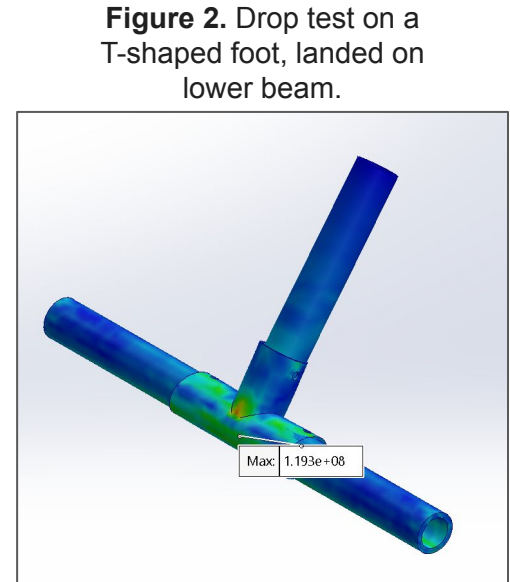


Figure 2. Drop test on a T-shaped foot, landed on lower beam.

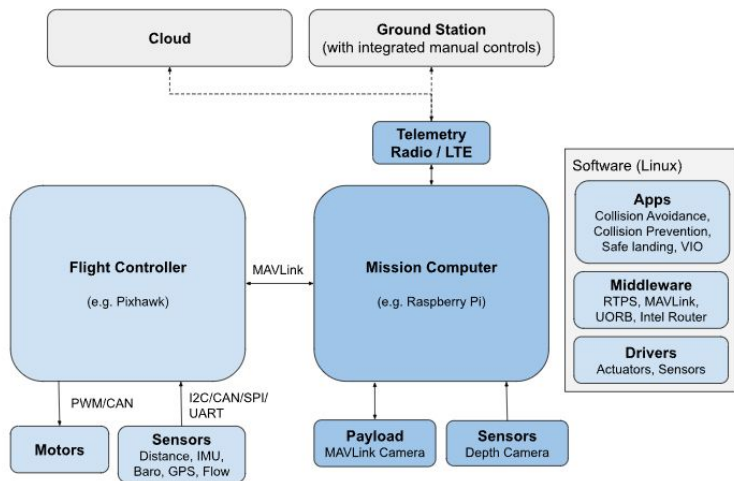
Both of these analyses were performed on ABS, which is generally weaker, but more elastic, than carbon fibre.

Autonomy

Autonomy



```
commander check - tells you if you are ready to fly/arm
commander status - gives current drone status and navigation mode
commander mode <mode_name> - sets the navigation mode
commander arm - arms drone
commander disarm - disarms drone
dmesg - prints PX4 log
commander help - prints the help menu for the commander commands
```



The software stack relies on many modules and libraries for flight control, data gathering, image processing, and environment visualization in 3D:

- A. **ROS2 Humble:** Open-Source Framework to handle communication between nodes.
- B. **Sensor:** Library to interface with the thermal camera to gather video feed via USB serial.
- C. **Open CV:** Computer Vision library used to create 'Depth Maps' from Stereo Images to generate the 3D point clouds.
- D. **Unity:** 3D point cloud visualization.
- E. **PX4:** the autopilot firmware that will control the drones movement and communication, interfaced with ROS2 for offboard control

- UAV autonomy is being achieved via PX4 firmware simulation and deployment, and is interfacing with PX4 via the uORB messaging API. The team demonstrated multi-agent Software-In-The-Loop (SITL) simulation through Gazebo, with sensor output being published to ROS 2 topics.

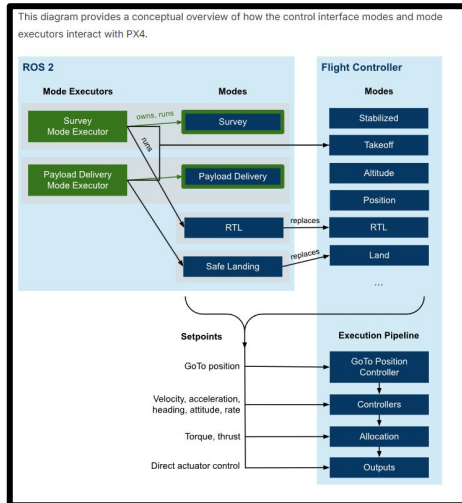


Figure: Multi-Drone simulation through Gazebo, using PX4



What is included in the electrical section depends on what phase you are in. Include some screenshots of wiring diagrams that needs to be discussed, but make sure to upload everything you have into the drive. Some examples of what to include are:

- Electronics diagrams
 - Level of detail depends on the phase. Research projects would just need a conceptual version, where prototyping versions should start having detailed diagrams that specify components
- Any calculations that support design choices
 - Battery calculations for example

Preliminary Autonomy System Requirements



ID	Description	Source	Derivation
A-R-1	The drones shall position themselves to collect useful data	PO-1	The drones have directional sensors and must point their sensors toward the target
A-R-2	The drones shall transmit telemetry data every 1s to a ground station	PO-3	The drone shall provide telemetry to validate autonomy at all times
A-R-3	The drones shall hover autonomously, and follow pre-set or adaptable trajectories.	PO-3	The drone should not require major manual input or intervention
A-R-4	In the event of catastrophic localization failure, the drone shall cut power	PO-4	The drone should prevent harm to persons or property

Visualization



What is included in the software section depends on what phase you are in. You can include things like block diagrams that highlight the logic of your code that needs to be discussed, but make sure to upload everything you have into the drive. Some examples of what to include are:

- Any pseudo code that has been written or planned
 - Mainly for Research Phase projects
- All written code should be in the TURTLE github
 - Mainly for Prototyping Phase projects
- Any logic diagrams or algorithms that need specific discussion in the DR should also be included
- Any edge cases or corner cases that should be considered or need discussion in the DR
 - Ex: code needs to account for motor configuration space or loss of a degree of freedom

Preliminary Visualization System Requirements



ID	Description	Source	Derivation
V-R-1	DRON shall transmit data continuously and in real time to a ground control station for visualization	PO-2	the data gathered should be relevant, intuitive, and recent
V-R-2	DRON shall update the environment with <10s of a real-time update	PO-2	the data gathered should be relevant, intuitive, and recent
V-R-3	DRON shall collect thermal and visual information about the environment	PO-1	thermal and visual information is useful to firefighters
V-R-4	DRON shall collect data every 10 seconds for the entire flight duration	PO-6	onboard storage should store data for the 10 min flight

Onboard Sensors

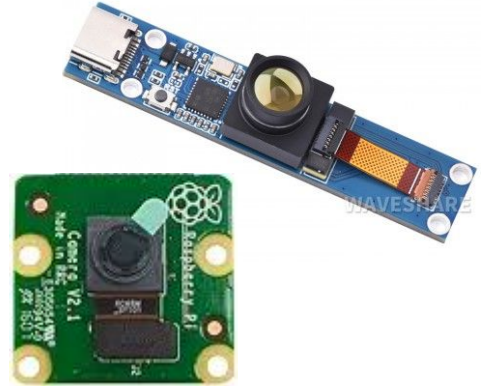


Cameras

- Long-wave IR Thermal Imaging Camera Module
- x2 Raspberry Pi Camera Module V2-8

Sensors

- IMU (on FC)
 - Barometer (on FC)
 - Micro M10 GPS
- > fused into state est. using EKF on FC



Computing

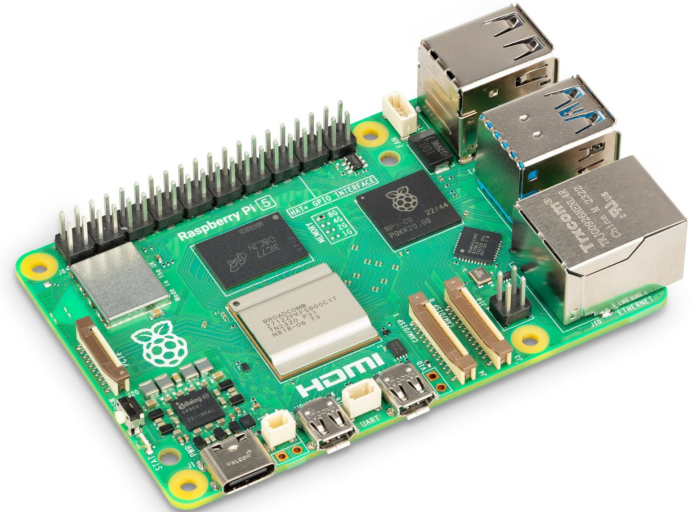


Raspberry PI 5

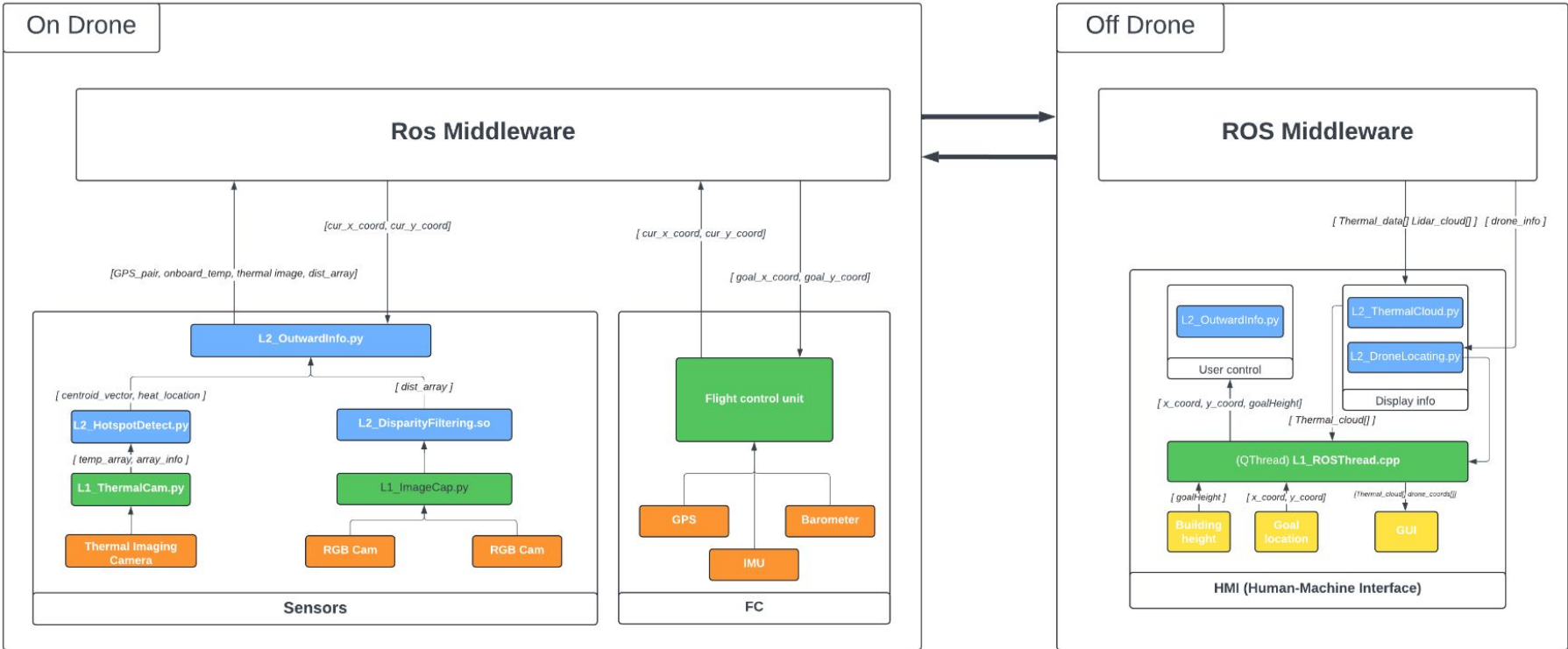
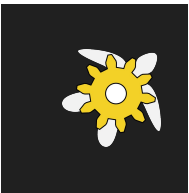
- 2 camera support
- Allows us to offboard some processing to the drones

Ground station (Laptop)

- Does more intense processing
- Allows the user to view the mappings



Processing loop



Dual Cameras



We take simultaneous images from the 2 onboard cameras



Image Rectification



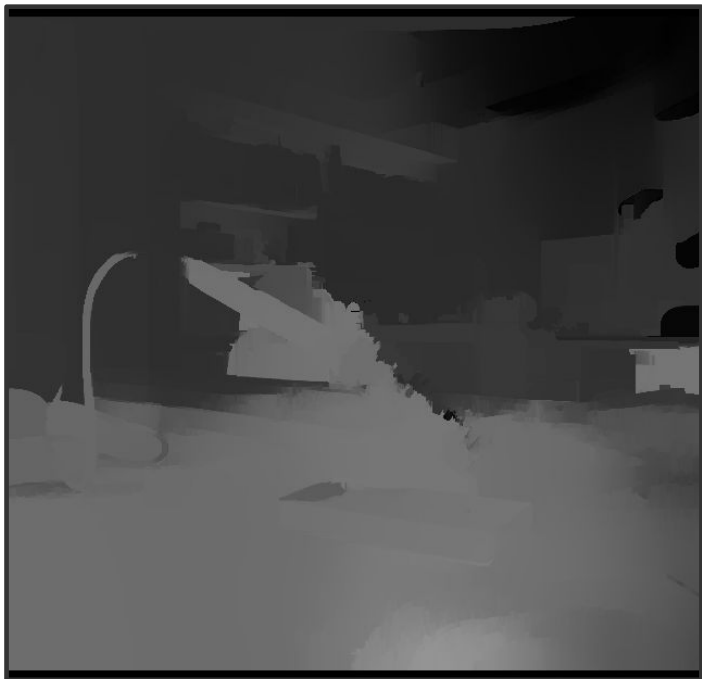
In order to reduce camera distortion and ensure the image elements overlap we rectify them



Depth map



We use OpenCV to take those images and convert them to a depth map



Voxel renderer

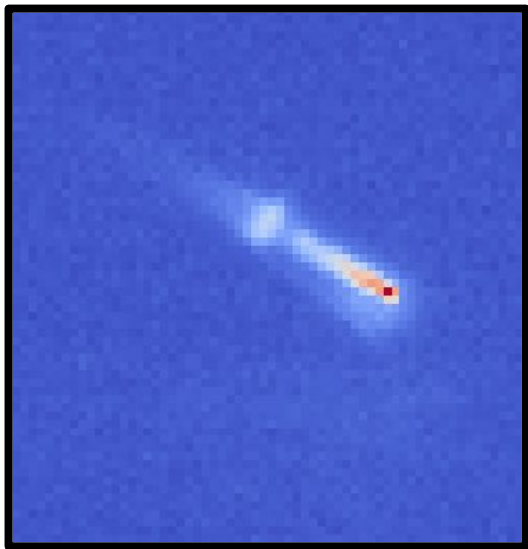


The data from the sensors is then rendered as a voxelized potential field in Open3D

Thermal vision



We use a Thermal USB Camera capture thermal data we



Combined image



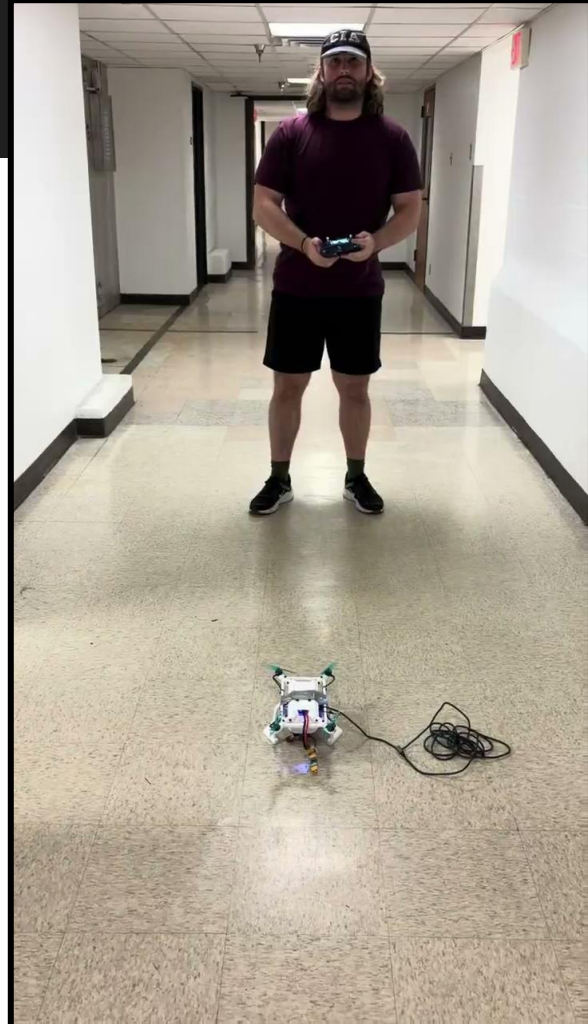
Field Testing

Flight test 3/20/2026



Wins:

- Demonstrated flight
- Validation of arms + legs



Flight test 3/22/2026 (1 of 2)



Wins:

- Sustained flight
- “Test Drone” design validated
 - Intended failure point on leg during hard landing
 - Reinforced arm design by frame
- Perturbation rejection (wind)



Flight test 3/22/2026 (2 of 2)



Timeline and BOM

Gantt Chart



Two areas of focus for the Gantt chart:

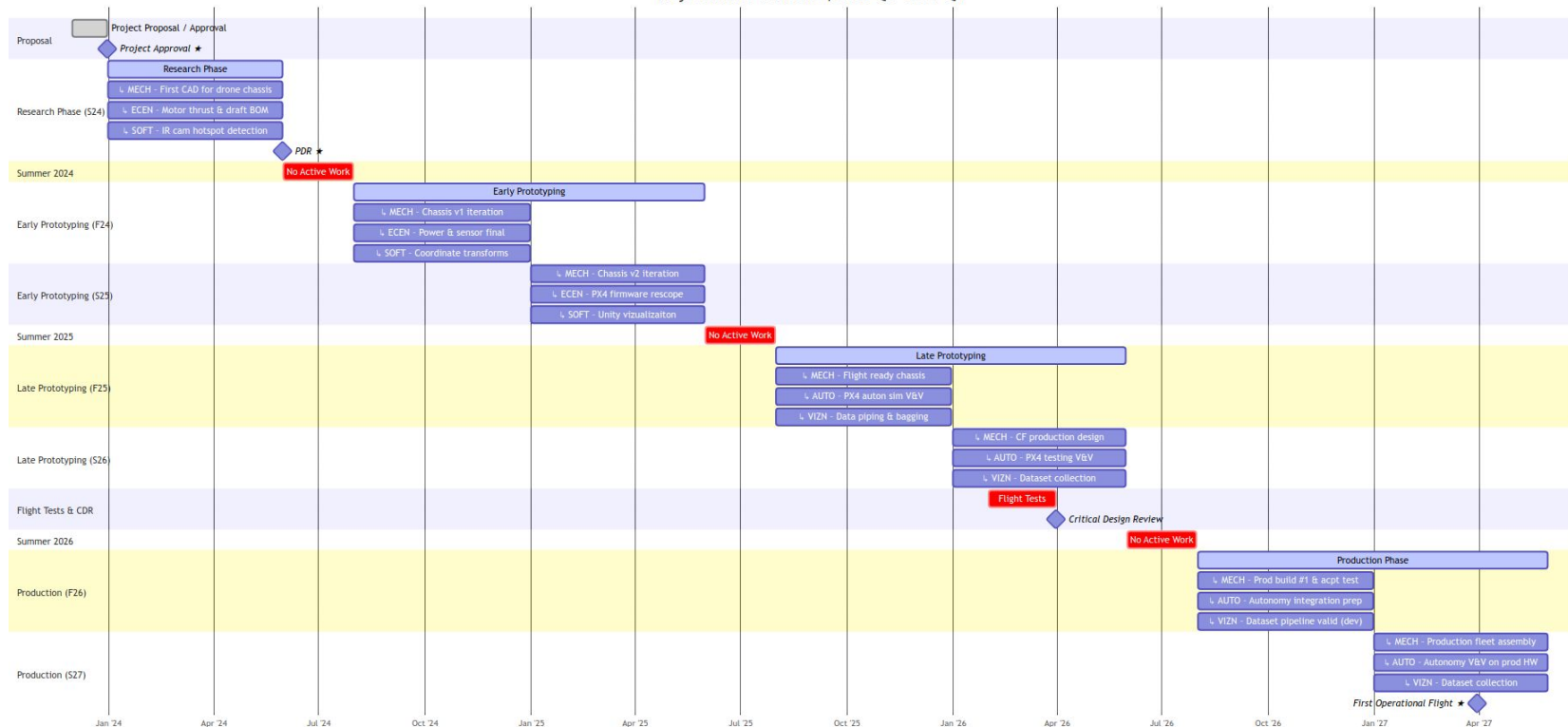
- 1) More detailed look at schedule for the semester to get your project to showcase
- 2) More general look at schedule for the lifetime of the project to get it to completion

The chart must contain the all necessary project tasks along with who is responsible for each task. An overview or simplified picture can be included in the slides, but a more detailed should be uploaded to the drive for review

Gantt Chart



Project Master Timeline | 2023 Q4 - 2027 Q2





- A list of materials, components, and the quantities of each needed to progress in their phase or continue to next phase.
- All purchases necessary until their next DR
 - For CDR, list all purchases for your one-time \$1000 budget
- There should be a detailed spreadsheet that has links to what you are looking to buy uploaded to the drive for review
- REMINDER - In the TURTLE Robotics Google Drive, if you go to Internal Affairs -> Treasury, in that folder you will find a spreadsheet labeled 'Bill of Materials Template', which is a great resource to use to make your BOM
 - For research and production projects, make sure to change the total budget to accurately reflect what you have access to

BOM



Table1_2									
Category	Item	Qty pe	Total Qty	Cost (\$)	Net Cost (\$)	Mass (g)	Net Mass (g)	Notes	
Electronics	GPS	1	2	\$25.00	\$53.50	2.0	2.0		
Electronics	Battery	1	2	\$33.00	\$70.62	205.0	205.0		
Electronics	rbg cams	2	4	\$14.00	\$59.92	3.0	6.0		
Electronics	Raspi 5	1	2	\$60.00	\$128.40	47.0	47.0	4 GB	
Electronics	IR cam	1	2	\$80.00	\$171.20	20.0	20.0	45 deg lens, ok	
Electronics	pi cam cables	2	4	\$10.00	\$42.80	0.5	1.0		
Electronics	Buck Converter	1	2	\$0.80	\$1.71	5.0	5.0		
Electronics	Reciever	1	2	\$23.00	\$49.22	5.5	5.5	2.4 ghz	
Electronics	smaller flash dr	1	2	\$19.00	\$40.66	0.0	0.0		
Electronics	FC + ESC	1	2	\$70.00	\$149.80		0.0		
Ground Station	Router	0	0	\$0.00	\$0.00	0.0	0.0	Aidan <3	
Ground Station	ELRS transmitt	1	1	\$80.00	\$85.60	0.0	9.1		
Mechanical	Motor	4	8	\$15.30	\$130.97	9.1	36.4	1404 3800 KV	
Mechanical	Props	4	8	\$1.00	\$8.56	1.5	5.8	T4x2	
Mechanical	Chassis	1	2	\$0.00	\$0.00	0.0	0.0	PLA is free	

Category	SUM of Net Cost (\$)
Electronics	\$767.83
Ground Station	\$85.60
Mechanical	\$139.53
Grand Total	\$992.96



TEXAS A&M UNIVERSITY
ROBOTICS TEAM & LEADERSHIP EXPERIENCE

Thank You, Questions?